Project Design Phase

Solution Architecture

Date	8 September 2026
Team ID	Individual
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Project Name	Digital Metro Ticket Booking System using ServiceNow
Maximum Marks	4 Marks

Solution Architecture:

The solution architecture for the Digital Metro Ticket Booking System bridges the metro ticketing problem with a ServiceNow-based digital solution. It provides a structured booking process in which passengers enter journey details, the system calculates the fare, and a digital ticket with a QR code is generated.

Its goals are to:

• Provide a standardized Service Catalog process for digital metro ticket booking.

• Capture Source, Destination, Passenger Type, and Number of Tickets through the booking form.

• Automate fare calculation and station details mapping.

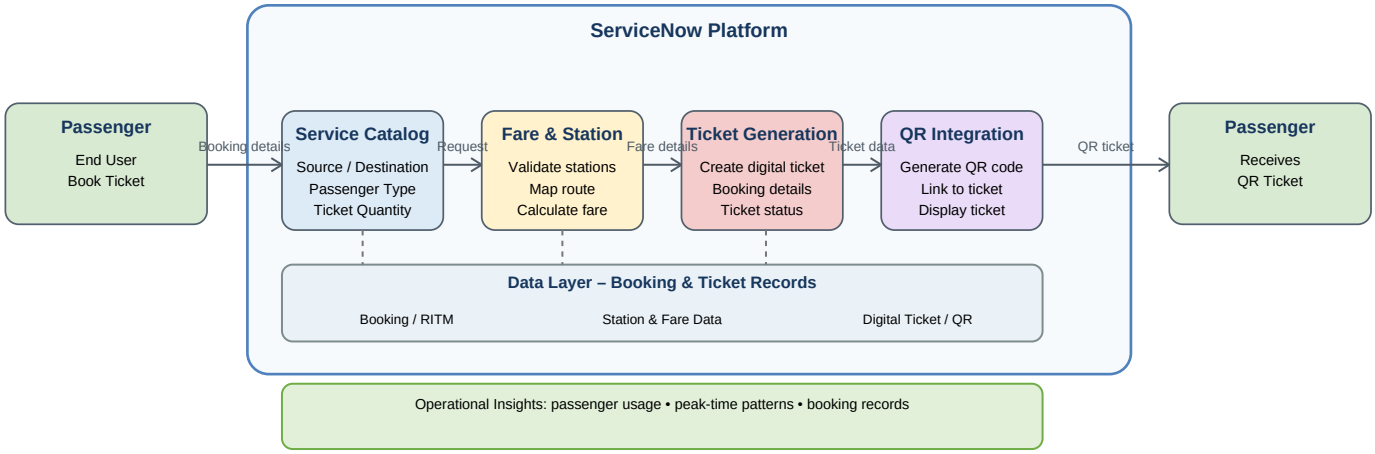
• Generate a digital ticket and QR code after successful booking.

• Maintain booking information for ticket tracking and operational analysis.

Project-specific architecture:

A passenger submits the Metro Ticket Booking catalog item by selecting the source, destination, passenger type, and number of tickets. ServiceNow processes the booking details and applies the configured station and fare logic. After the fare is calculated, the system generates a digital ticket and QR code. The ticket details can then be displayed to the passenger through the configured ServiceNow interface. The stored booking data can also support operational insights such as passenger usage and peak-time patterns.

Figure 1: Solution architecture and data flow of the Digital Metro Ticket Booking System



Architecture Components:

Component	Purpose
Passenger / Requestor	Initiates the digital metro ticket booking request and provides journey and passenger details.
Service Catalog	Provides the standardized Metro Ticket Booking item and captures Source, Destination, Passenger Type, and Number of Tickets.
ServiceNow Data Layer	Stores booking and ticket information required for processing and tracking.
Fare Calculation Logic	Calculates the applicable fare using the configured station, route, passenger, and ticket quantity information.
Station Details Mapping	Maintains and uses station information for source/destination selection and route-based fare processing.
Workflow / Flow Designer	Coordinates the booking workflow and automation between submission, processing, ticket generation, and status updates.
Digital Ticket Generation	Creates the digital ticket containing journey, passenger, quantity, and fare details.
QR Code Integration	Generates the QR code associated with the digital ticket for convenient ticket access and validation.
Ticket Tracking	Maintains booking/ticket information so the passenger and authorized teams can reference the ticket status and details.
Operations & Analytics	Uses digital booking information to support analysis of passenger behavior, peak usage times, and operational patterns.

Architecture Flow Summary:

Passenger → **Service Catalog** → **Station Details & Fare Calculation** → **Digital Ticket Generation** → **QR Code Integration** → **Ticket Access.**

Implementation Alignment

The architecture directly supports the planned execution roadmap: Catalog Form Setup → Fare Logic Script → QR Code Integration → Station Details Mapping → UI Pages → UAT & Deployment. The stakeholders supported by the solution are Passengers, Station Managers, the Metro Operations Team, and IT Administrators.

Key Solution Benefits

The architecture supports faster digital ticket booking, reduced manual ticketing effort, automated fare processing, paperless QR-based tickets, and structured booking data that can support operational decision-making.